Architecture of the Global weather Dashboard

Project Overview The ClimateScope dashboard is designed to analyze global weather patterns using the Global Weather Repository dataset. It helps identify climate trends, correlations between weather variables, extreme weather events, and regional climate variability through interactive visualizations.

Dataset Description:

Key Variables: -

temperature_celsius

humidity

precip_mm

wind_kph

pressure_mb

location_name

country

Global weather Dashboard

Filters:

Country Climate Zone Region Type Hemisphere

Year Month Date Range Reset All Filters

Kpi summary section:

Avg Global Temp Avg Humidity Total Rainfall

Avg Wind Speed Extreme Weather Events

global weather map:

Temperature Humidity Rainfall Wind Patterns pressure etc….

climate trend analysis**:**

Temperature Trend Rainfall Trend Humidity Trend Wind Speed Trend

regional comparison

Top Hottest Countries Highest Rainfall Regions Highest Humidity Areas
Strongest Wind Regions

extreme weather analysis:

Heatwaves Storm Events Compound Extremes (High Heat + High Wind)

Extreme Temperature Days

A. Filters (Slicers)

Used to dynamically update all visualizations on the dashboard.

- **Geographic:** Country, Climate Zone, Region Type, Hemisphere.
- **Temporal:** Year, Month, Date Range.
- **Action: Reset All Filters** button to clear selections.

B. KPI Summary Section (Quick Insights)

Provides aggregate values for immediate performance tracking.

- **Avg Global Temp:** Mean of temperature_celsius.
- **Avg Humidity:** Mean of humidity.
- **Total Rainfall:** Sum of precip_mm.
- **Avg Wind Speed:** Mean of wind_kph.
- **Extreme Weather Events:** Count of records where extreme_event is flagged.

C. Global Weather Map (Main Visualization)

Different type of maps displaying intensity levels for:

- Temperature, Humidity, Rainfall, Wind Patterns, and Atmospheric Pressure.

D. Climate Trend Analysis

Time-series line charts showing historical movement.

- **Variables:** Temperature Trend, Rainfall Trend, Humidity Trend, Wind Speed Trend.

E. Regional Comparison

Ranked visualizations (Bar Charts/Treemaps) highlighting top-performing areas.

- **Metrics:** Top Hottest Countries, Highest Rainfall Regions, Highest Humidity Areas, Strongest Wind Regions.

F. Extreme Weather Analysis

Deep-dive and risk factors.

- **Heatwaves:** Frequency of extreme_temperature_days.
- **Storm Events:** High wind_kph and pressure_change correlation.
- **Compound Extremes:** Analysis of overlapping events (e.g., High Heat + High Wind).